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(54) **DOOR STOP AND DOOR FRAME ASSEMBLY**

(71) Applicant: **SHIEN CHENG ENTERPRISE CO., LTD.**, Taichung (TW)

(72) Inventor: **Keng-Hao Chang**, Taichung (TW)

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E05F 5/06 (2006.01)

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(52) **U.S. Cl.**

CPC **E06B 1/52** (2013.01); **E05F 5/06** (2013.01); **E06B 1/045** (2013.01)

(58) **Field of Classification Search**

CPC E05F 5/06; E06B 1/52; E06B 1/526; E06B 1/18; E06B 1/30; E06B 1/045
See application file for complete search history.

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Primary Examiner — Justin B Rephann

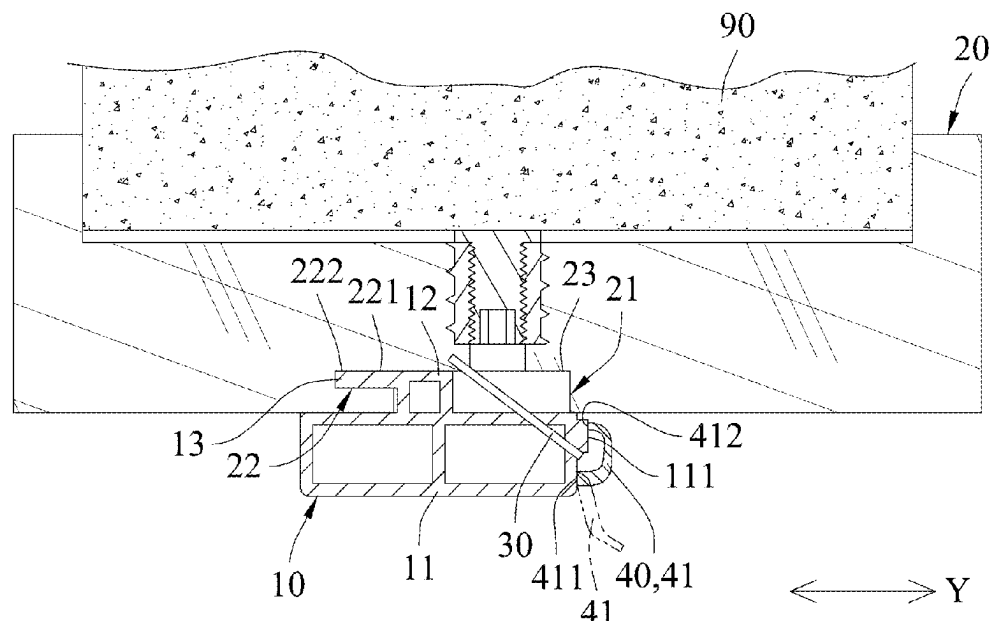
(74) Attorney, Agent, or Firm — Bruce Stone LLP;
Joseph A. Bruce

(57)

ABSTRACT

A door stop and door frame assembly to be mounted on a wall, includes: a door stop strip in an elongated strip shape, and including a body and a connecting portion located on one side of the body and an insertion portion connected to the connecting portion, wherein one end of the body is an assembling end; a door frame mounted on the wall, wherein one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the connecting portion is located in the accommodation groove, and the insertion portion is located in the insertion groove; a fixing member inserted into the door frame and the door stop strip to fix the door stop strip to the door frame; and the assembling end is connected to a buffer, and the buffer is pushed against a door panel.

8 Claims, 8 Drawing Sheets



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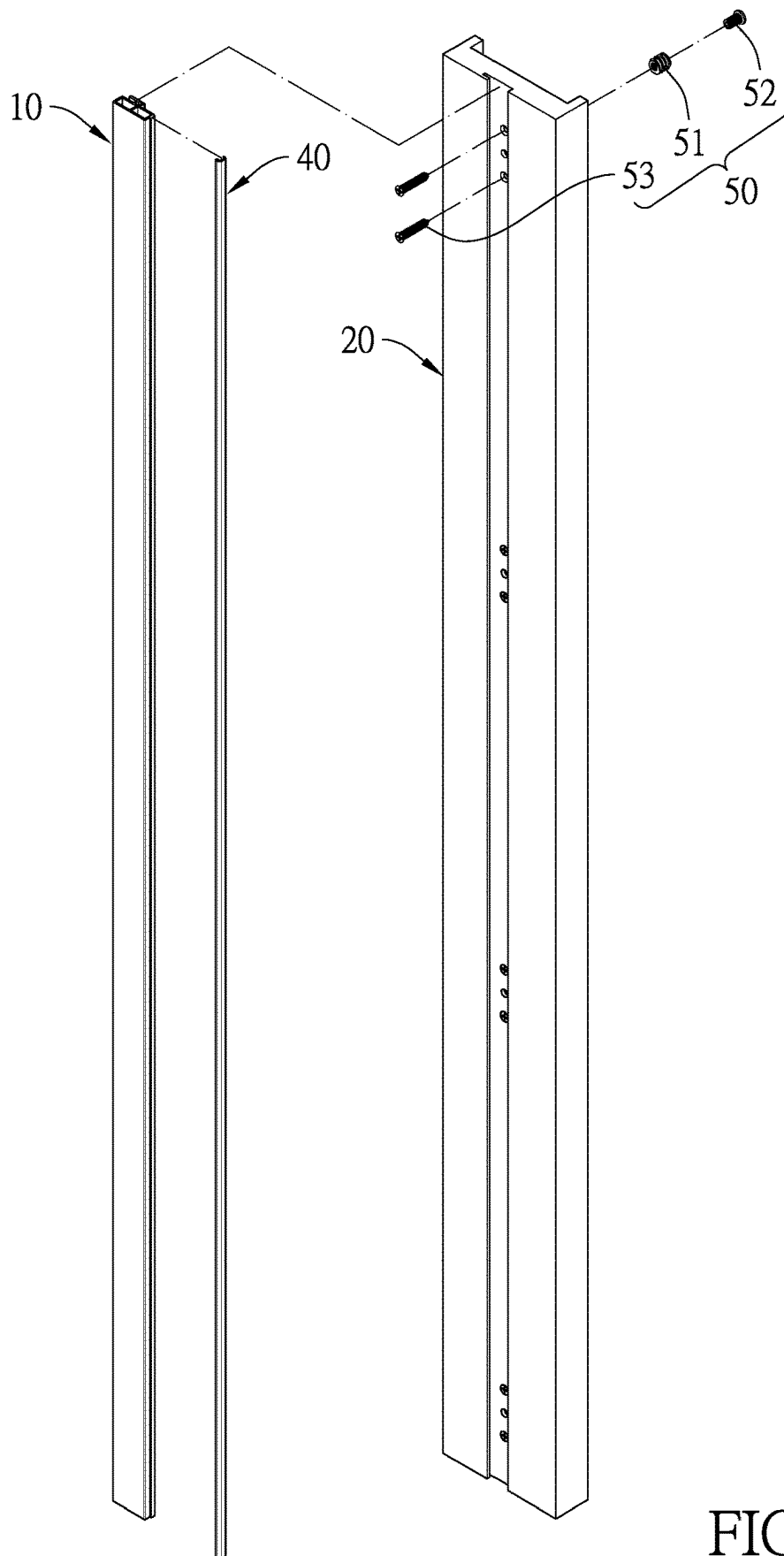


FIG.1

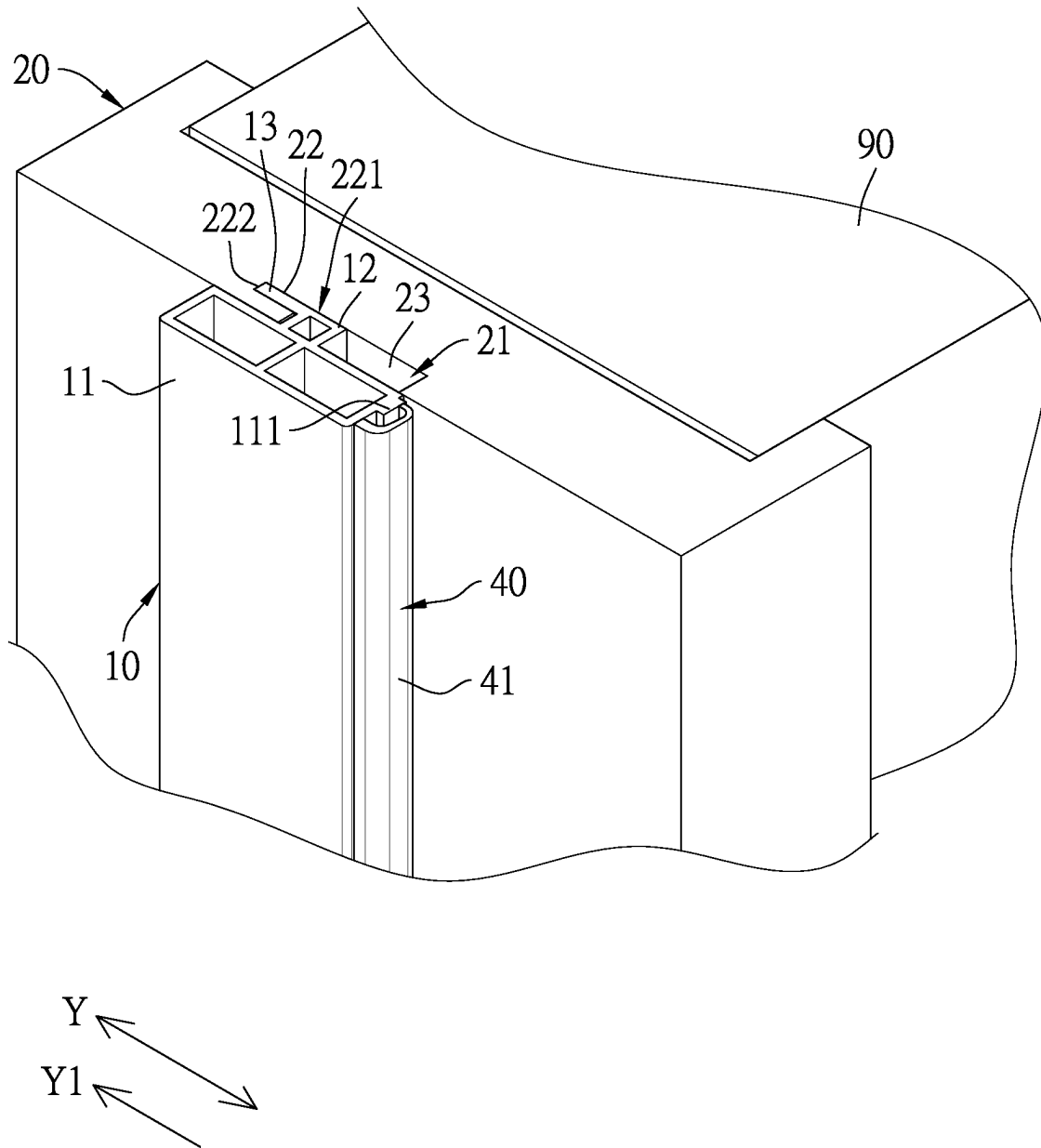


FIG.2

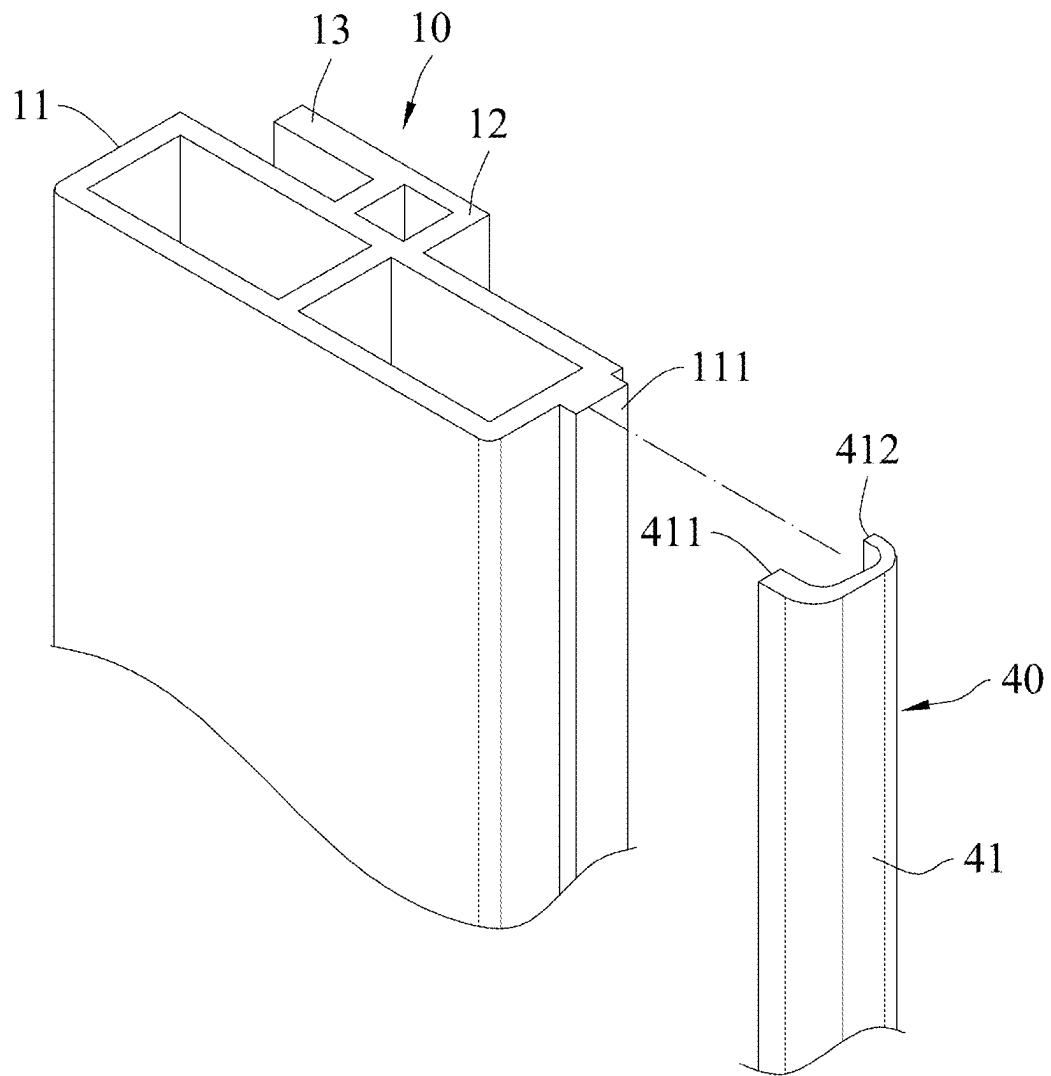


FIG.3

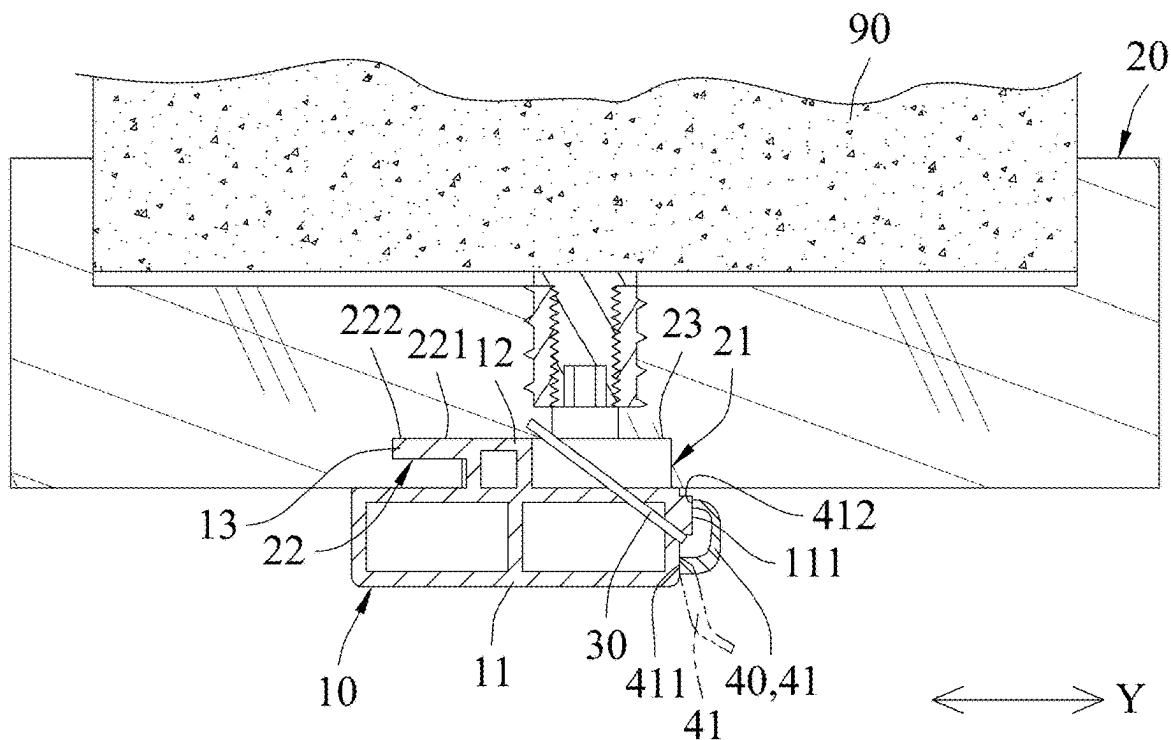


FIG. 4

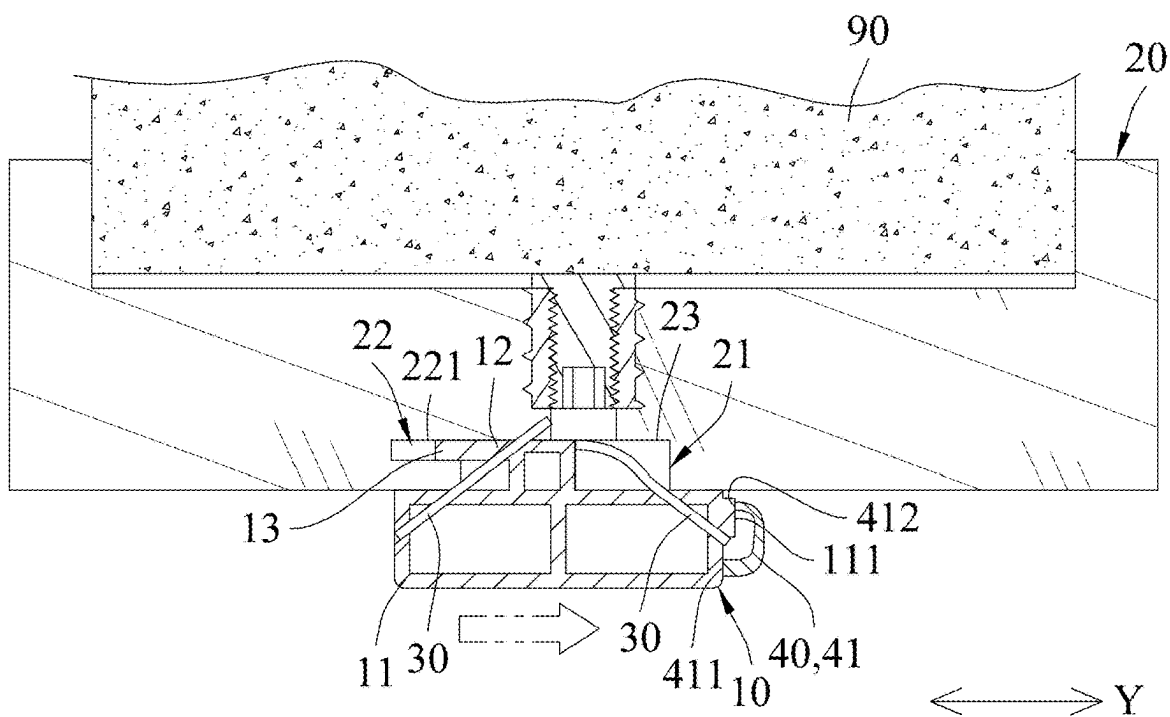


FIG. 5

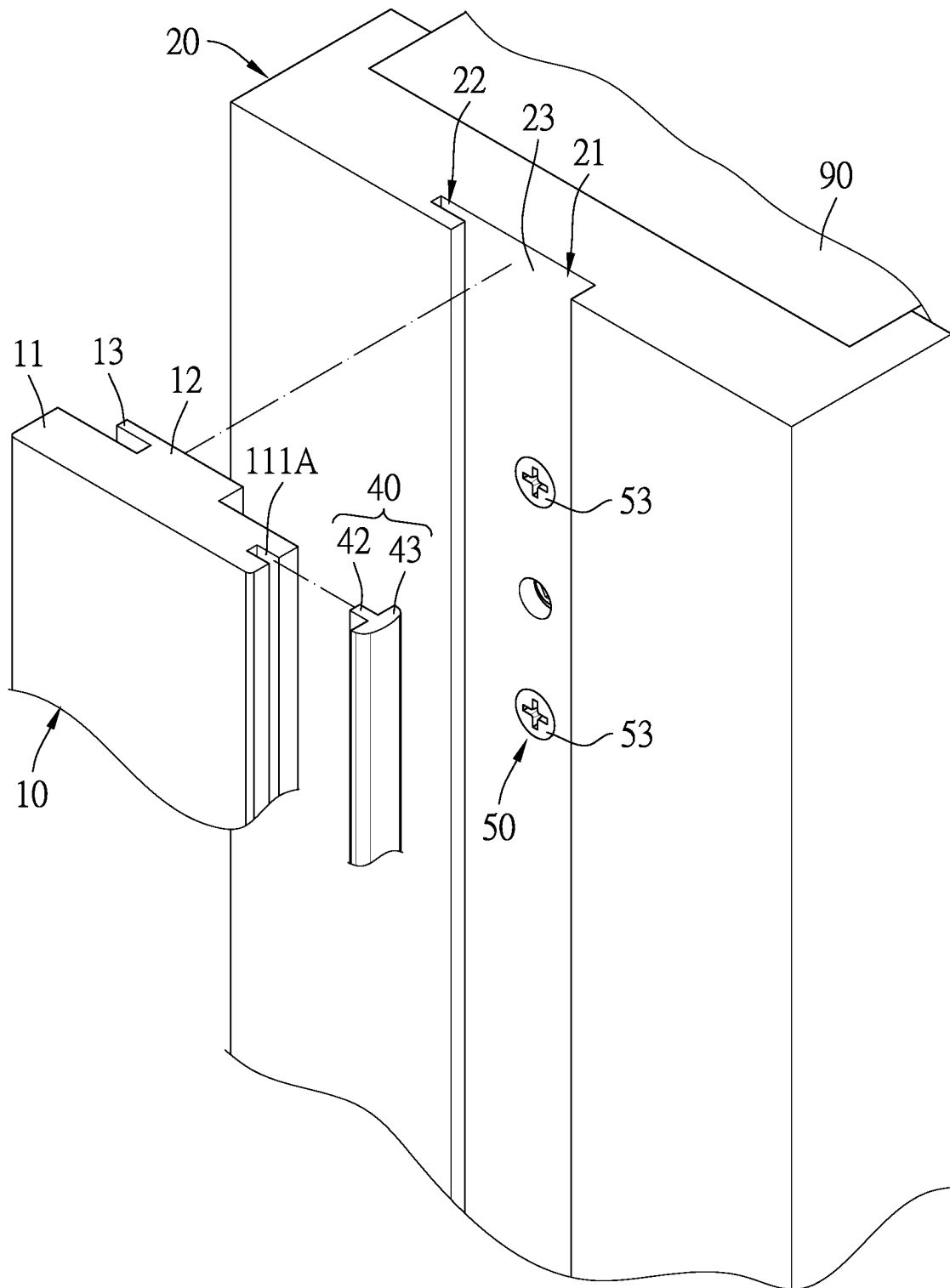


FIG.6

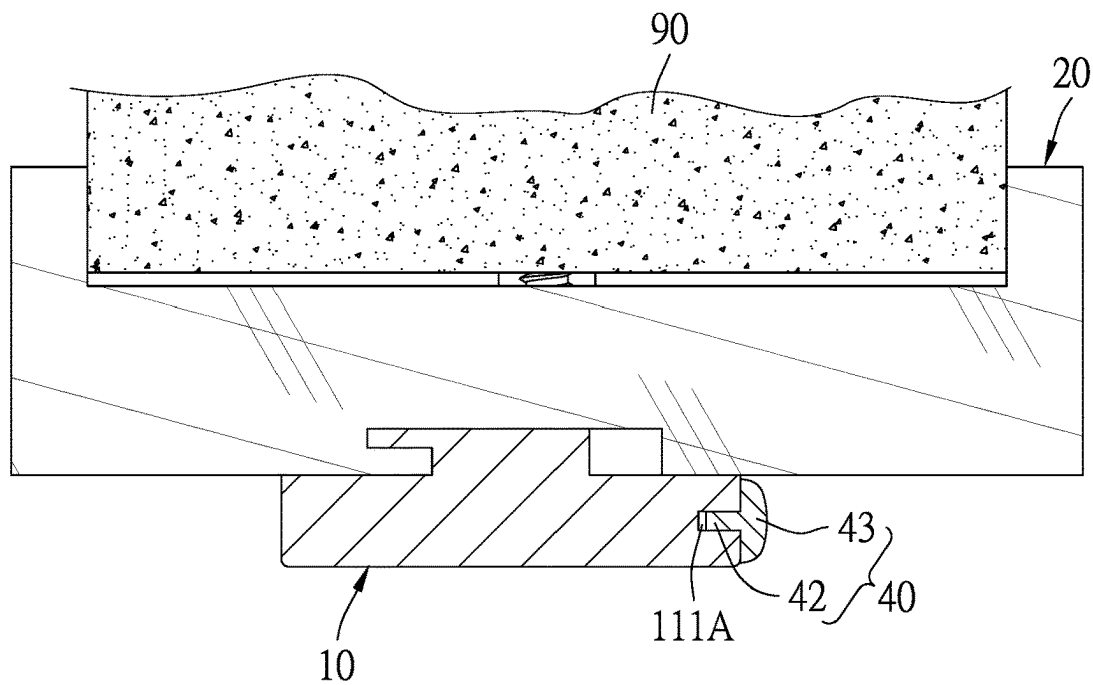


FIG. 7

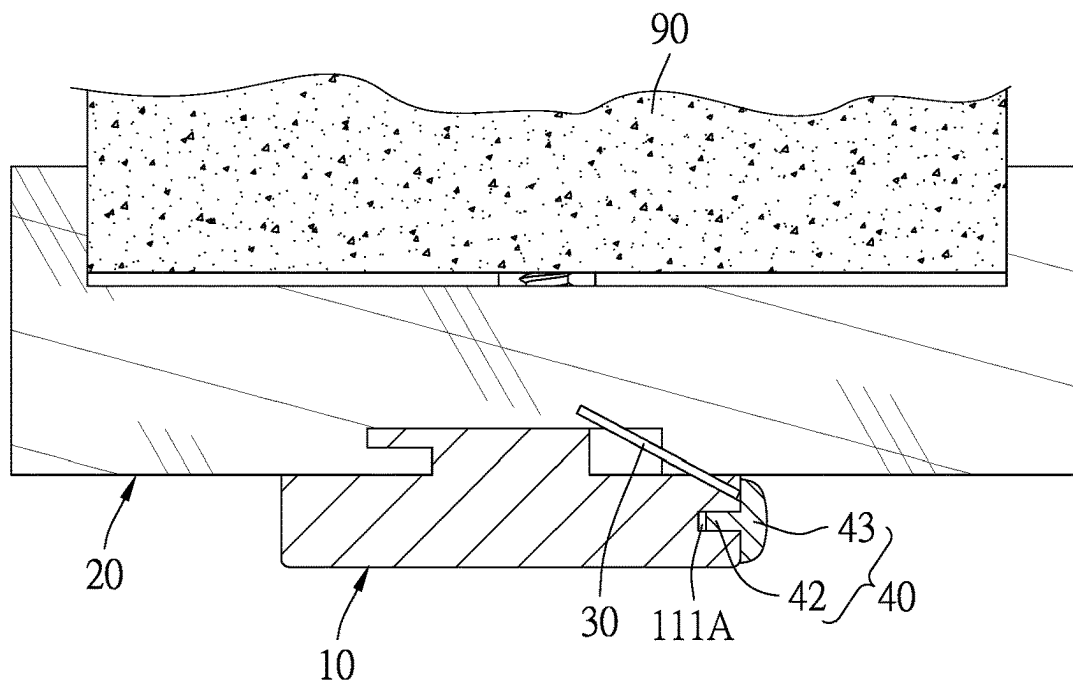


FIG. 8

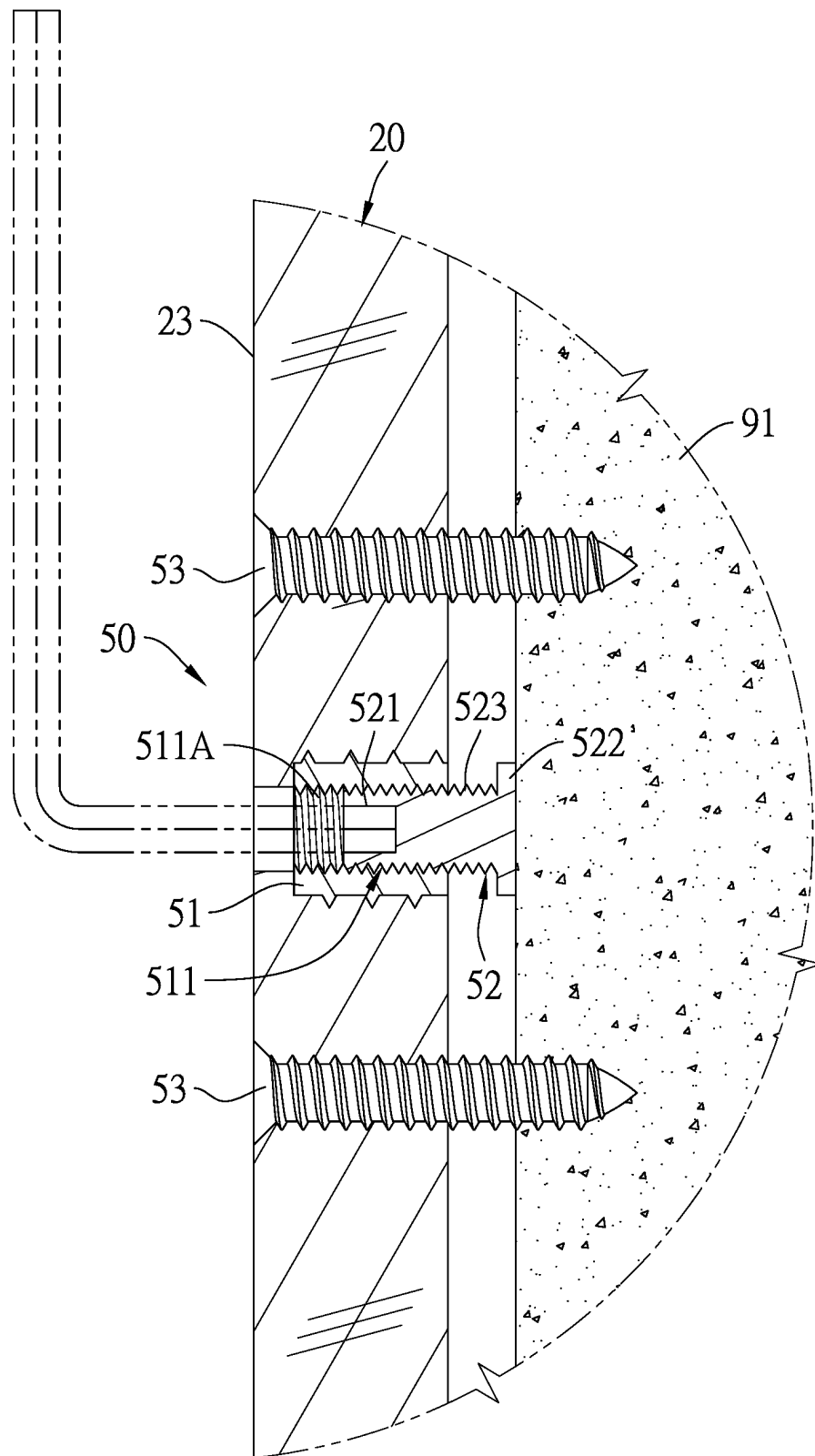


FIG.9

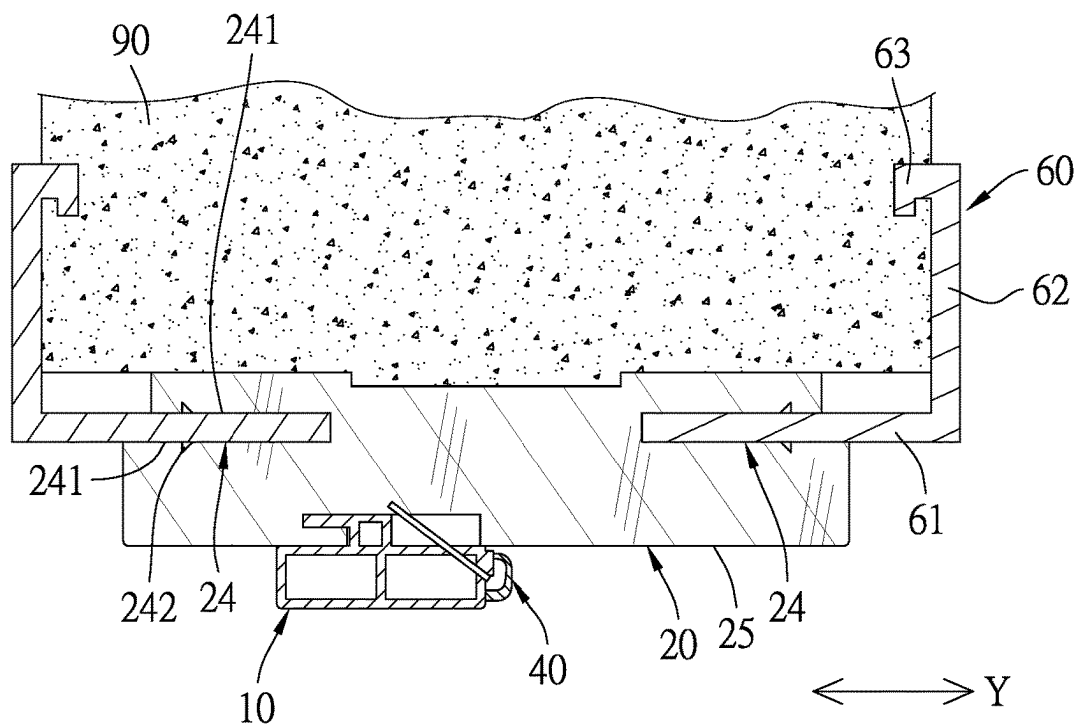


FIG. 10

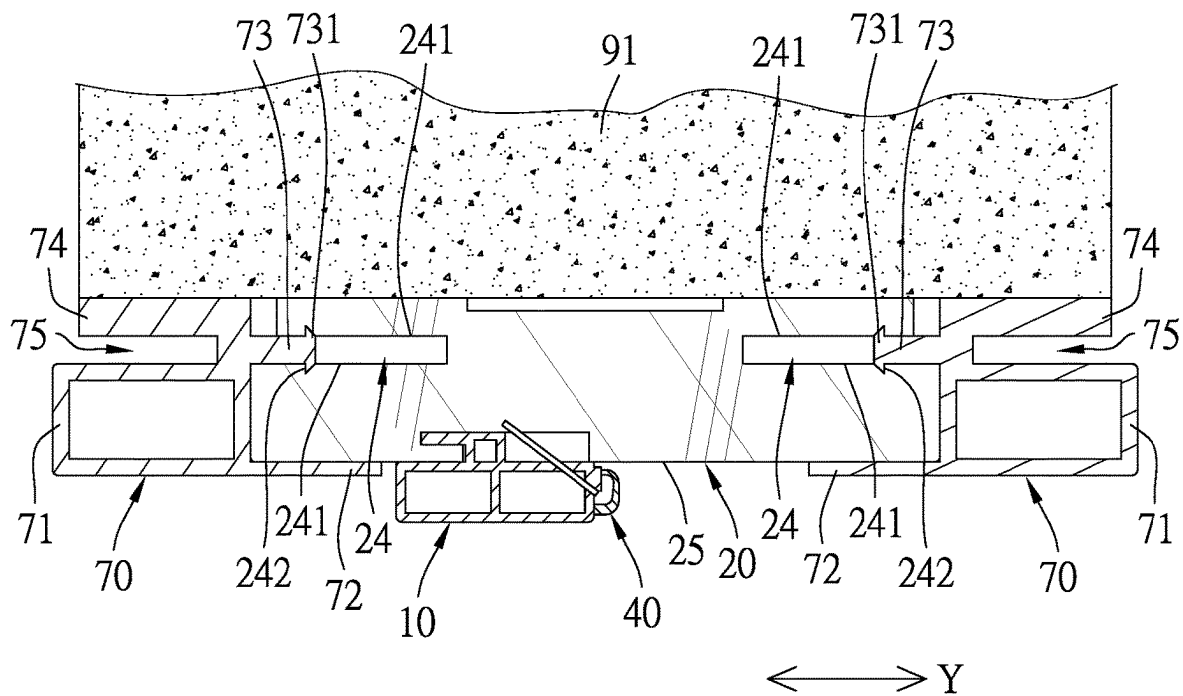


FIG. 11

1

DOOR STOP AND DOOR FRAME ASSEMBLY

BACKGROUND

Field of the Invention

The present invention relates to a door stop, and more particularly to a door stop and door frame assembly.

Description of Related Art

Generally, the door stop is used to stop the door frame, and the door stop and the door frame can be formed as one or two pieces of structure, when the door stop and door frame are two-piece, the door stop and door frame are in a concave and convex fit, combined in a mutual locking manner, so that although the door stop can cover the nail heads locked on the door frame, the door stop can not fine tune its position. When the door is used for a long time, the stability of the fit between the door stop and the door panel will be greatly reduced, if the position of the door stop can not be adjusted, it will lead to the door panel can not be correctly against the door stop, which may cause problems such as unreliable closing. Therefore, a door stop and door frame assembly that can adjust the position of the door stop is needed.

SUMMARY

The present invention provides a door stop and door frame assembly, the main objective of which is to enable the door stop to be shifted relative to the door frame and fine-tuned so that the door panel is correctly pressed against the door stop.

To achieve the above objective, a door stop and door frame assembly configured to be mounted on a wall, comprises:

- a door stop strip in an elongated strip shape, and including a body and a connecting portion located on one side of the body and an insertion portion connected to the connecting portion, wherein one end of the body is an assembling end;
- a door frame mounted on the wall, wherein one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the connecting portion is located in the accommodation groove, and the insertion portion is located in the insertion groove;
- a fixing member inserted into the door frame and the door stop strip to fix the door stop strip to the door frame; and the assembling end is connected to a buffer, and the buffer is pushed against a door panel.

As can be seen from the foregoing, the present invention mainly provides the door stop strip with the insertion portion and the connecting portion, one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the connecting portion is located in the accommodation groove, the insertion portion is located in the insertion groove, the insertion groove and the accommodation groove are both in an elongated strip shape, so that the position of the insertion portion in the insertion groove can be fine-adjusted, then the position of the door stop strip relative to the door frame can be adjusted, so that the door stop can be shifted relative to the door frame and fine-tuned, and thus the door panel can be correctly pressed against the door stop.

2

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an exploded view of a door stop and door frame assembly in accordance with the present invention;

FIG. 2 is an illustrative view of the door stop and door frame assembly in accordance with the present invention;

FIG. 3 is an exploded view of a door stop in accordance with the present invention;

FIG. 4 is an assembly view of the door stop and door frame assembly in accordance with the present invention;

FIG. 5 is an illustrative view of the door stop and door frame assembly in accordance with the present invention;

FIG. 6 is an exploded view of the door stop and door frame assembly in accordance with the present invention;

FIG. 7 is an illustrative view of the door stop and door frame assembly in accordance with the present invention;

FIG. 8 is an illustrative view of the door stop and door frame assembly in accordance with the present invention;

FIG. 9 is an illustrative view of the adjustment module in accordance with the present invention;

FIG. 10 is an illustrative view of the fasteners fixed to the door frame assembly in accordance with the present invention; and

FIG. 11 is an illustrative view of the expansion members fixed to the door frame assembly in accordance with the present invention.

DETAILED DESCRIPTION

The present invention provides a door stop and door frame assembly, as shown in FIG. 1-9, including:

A door stop strip **10** is an elongated strip, and includes a body **11** and a connecting portion **12** located on one side of the body **11** and an insertion portion **13** having one end connected to the connecting portion **12**, the insertion portion **13** extends along a transverse direction Y and is in an elongated strip shape, and one end of the body **11** along the transverse direction Y is an assembling end **111**;

A door frame **20** is mounted on a wall **90**, and one side of the door frame **20** facing away from wall **90** includes an accommodation groove **21** and an insertion groove **22** that are connected along the transverse direction Y. The insertion groove **22** and the accommodation groove **21** extend along the transverse direction Y and are in an elongated strip shape. The connecting portion **12** is located in the accommodation groove **21**, and the insertion portion **13** is located in the insertion groove **22**;

Preferably, one end of the insertion groove **22** includes an opening **221**, and the other end of the insertion groove **22** has a bottom **222**. The opening **221** is connected to the accommodation groove **21**, and the direction from the opening **221** to the bottom **222** is an impact direction Y1. It is worth mentioning that the transverse direction Y is bidirectional, and the impact direction Y1 is one of the transverse direction Y. One end of the insertion portion **13** that is not connected to the connecting portion **12** is rested against the bottom **222**, thereby, when the door panel hits the door stop strip **10**, it will make the door stop strip **10** more closely fit with the door frame **20**.

A fixing member **30** is inserted into the door frame **20** and the door stop strip **10** to fix the door stop strip **10** to the door frame **20**. In this embodiment, the fixing member **30** is

3

inserted into the door frame **20** in a manner tilted from the assembling end **111**, and the fixing member **30** is a mosquito nail;

In a preferred embodiment, the assembling end **111** is connected to a buffer **40**, which is used to push against a door panel and provide buffering force for the door panel.

In this embodiment, please refer to FIGS. **4** and **5**, the buffer **40** is an arc-shaped member **41**, which is made of flexible materials, such as but not limited to PVC material. The arc-shaped member **41** includes a first arc-shaped end **411** and a second arc-shaped end **412** which are opposite to each other. The first arc-shaped end **411** of the arc-shaped member **41** is fixedly connected to the assembling end **111**, and the second arc-shaped end **412** of the arc-shaped member **41** is against the assembling end **111**, which can be pulled by the user to lift the arc-shaped member **41**. Furthermore, the arc-shaped member **41** can also cover the fixing member **30**, so that the user cannot see it.

In another embodiment, please refer to FIGS. **6-8**, the assembling end **111** further includes a groove **111A**, and the buffer **40** includes a fixing portion **42** and a buffering portion **43** that are connected with each other. The fixing portion **42** is inserted in the groove **111A** to fix the buffering portion **43** to the assembling end **111**, and the buffering portion **43** is abutted against the door panel.

In other embodiments, please refer to FIGS. **1**, **6**, and **9**, which also include an adjustment module **50**, including an embedded member **51** and an adjustment member **52**. The embedded member **51** is embedded in the door frame **20** and includes an inner threaded hole **511**. The inner threaded hole **511** includes a plurality of inner threads **511A**. The adjustment member **52** includes a wrench hole **521** at one end and a support portion **522** at the other. The adjustment member **52** further includes outer threads **523** on the peripheral wall thereof, and is disposed in the inner threaded hole **511** where the outer threads **523** are engaged with the inner threads **511A**. The support portion **522** is used to lean against the wall **90** or away from the wall **90**, and the wrench hole **521** allows the wrench to pass through to rotate the adjustment member **52**, thereby driving the support portion **522** to lean against the wall **90** or away from the wall **90**, so that the door frame **20** is pressed against or spaced at a distance from the wall **90**, in order to achieve the effect of fine tuning the position of the door frame **20**.

Preferably, the adjustment module **50** further includes a plurality of screws **53**, which are inserted through the door frame **20** and threaded into the wall **90** to secure the door frame **20** to the wall **90**.

Preferably, please refer to FIG. **6**, the door frame **20** includes a locking surface **23**, and the adjustment module **50** is exposed to the locking surface **23**. The locking surface **23** is one of the inner walls of the accommodation groove **21**, and the door stop strip **10** covers the locking surface **23** to prevent users from directly seeing the adjustment module **50**, in order to enhance the overall aesthetics of the door stop and door frame assembly in this invention.

In other embodiments of this invention, please refer to FIG. **10**:

The door frame **20** includes two elongated grooves **24** and two fasteners **60** which are respectively at two opposite ends thereof along the transverse direction **Y** (there are one elongated groove **24** and one fastener **60** at one end of the door frame **20**, and one elongated groove **24** and one fastener **60** at the other). The fasteners **60** each include a first section **61**, a second section **62**, and a hook **63** which are connected in sequence. The extension directions of the first section **61** and the second section **62** are different. In this

4

embodiment, the extension directions of the first section **61** and the second section **62** are perpendicular to each other. The first section **61** is inserted into the elongated groove **24** to fix the two fasteners **60** on the two opposite sides of the door frame **20**, the extension direction of the second section **62** is the same as that of the wall **90**, and the hook **63** is hooked to the wall **90** to fix the door frame **20** to the wall **90**.

In another embodiment of this invention, please refer to FIG. **11**:

the invention further includes two expansion members **70**, each of which includes a main body **71** and a supporting section **72**, a tenon section **73**, and a side section **74** that extend from the main body **71**. The supporting section **72** extends in the same direction as the tenon section **73**, and the side section **74** extends in the opposite direction to the supporting section **72** and the tenon section **73**. The supporting section **72** is abutted against an outer lateral surface **25** of the door frame **20**. One end of the supporting section **72** that is not connected to the main body **71** is abutted against one end of the door stop strip **10** that is faced away from the assembling end **111**. One end of the tenon section **73** that is not connected to the main body **71** includes a tenon convex portion **731**. Two opposite outer surfaces **241** of the elongated groove **24** each include an engaging groove **242**. The tenon section **73** extends into the elongated groove **24**, and the tenon convex portions **731** are engaged in the engaging grooves **242**, so that the expansion members **70** are locked to two opposite sides of the door frame **20**. An expansion groove **75** is formed between the side section **74** and the main body **71**, and the first section **61** of the fasteners **60** is inserted in the expansion groove **75**, so that the expansion members **70** of this embodiment can be fixed to two opposite sides of the door frame **20**, and can be adapted to walls **90** of different widths to effectively cover the end face of the wall **90**.

What is claimed is:

1. A door stop and door frame assembly configured to be mounted on a wall, comprising:

a door stop strip in an elongated strip shape, and including a body and a connecting portion located on one side of the body and an insertion portion connected to the connecting portion, wherein one end of the body is an assembling end, and the insertion portion is in an elongated strip shape;

a door frame mounted on the wall, wherein one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the insertion groove is in an elongated strip shape, the connecting portion is located in the accommodation groove, and the insertion portion is located in the insertion groove;

a fixing member inserted into the door frame and the door stop strip to fix the door stop strip to the door frame; the assembling end is connected to a buffer, and the buffer is pushed against a door panel; and

an adjustment module including an embedded member and an adjustment member, wherein the embedded member is embedded in the door frame and includes an inner threaded hole, the inner threaded hole includes a plurality of inner threads, the adjustment member includes a wrench hole at one end and a support portion at the other, the adjustment member includes outer threads and is disposed in the inner threaded hole where the outer threads are engaged with the inner threads.

5

2. The door stop and door frame assembly as claimed in claim 1, wherein the insertion portion extends along a transverse direction, one end of the body along the transverse direction is an assembling end, the accommodation groove and the insertion groove are connected along the transverse direction, and the accommodation groove is in an elongated strip shape.

3. The door stop and door frame assembly as claimed in claim 1, wherein the buffer is an arc-shaped member and made of flexible materials, the arc-shaped member includes a first arc-shaped end and a second arc-shaped end which are opposite to each other, the first arc-shaped end of the arc-shaped member is fixedly connected to the assembling end, and the second arc-shaped end of the arc-shaped member is against the assembling end, and the arc-shaped member covers the fixing member.

4. The door stop and door frame assembly as claimed in claim 1, wherein the assembling end further includes a groove, and the buffer includes a fixing portion and a buffering portion that are connected with each other, the fixing portion is inserted in the groove to fix the buffering portion to the assembling end, and the buffering portion is abutted against the door panel.

5. The door stop and door frame assembly as claimed in claim 1, wherein the adjustment module further includes a plurality of screws, which are inserted through the door frame.

6. The door stop and door frame assembly as claimed in claim 1, wherein the door frame includes a locking surface, the adjustment module is exposed to the locking surface, and the door stop strip covers the locking surface.

7. A door stop and door frame assembly configured to be mounted on a wall, comprising:

a door stop strip in an elongated strip shape, and including a body and a connecting portion located on one side of the body and an insertion portion connected to the connecting portion, wherein one end of the body is an assembling end, and the insertion portion is in an elongated strip shape;

a door frame mounted on the wall, wherein one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the insertion groove is in an elongated strip shape, the connecting portion is located in the accommodation groove, and the insertion portion is located in the insertion groove; and

a fixing member inserted into the door frame and the door stop strip to fix the door stop strip to the door frame; wherein the assembling end is connected to a buffer, and the buffer is pushed against a door panel;

the insertion portion extends along a transverse direction, one end of the body along the transverse direction is an assembling end, the accommodation groove and the insertion groove are connected along the transverse direction, and the accommodation groove is in an elongated strip shape;

wherein the door frame includes two elongated grooves and two fasteners which are respectively at two opposite ends thereof along the transverse direction, the

6

fasteners each include a first section, a second section, and a hook which are connected in sequence, extension directions of the first section and the second section are different, the first section is inserted into the elongated groove to fix the two fasteners on the two opposite sides of the door frame, and the hook is hooked to the wall.

8. A door stop and door frame assembly configured to be mounted on a wall, comprising:

a door stop strip in an elongated strip shape, and including a body and a connecting portion located on one side of the body and an insertion portion connected to the connecting portion, wherein one end of the body is an assembling end, and the insertion portion is in an elongated strip shape;

a door frame mounted on the wall, wherein one side of the door frame facing away from wall includes an accommodation groove and an insertion groove that are connected with each other, the insertion groove is in an elongated strip shape, the connecting portion is located in the accommodation groove, and the insertion portion is located in the insertion groove;

a fixing member inserted into the door frame and the door stop strip to fix the door stop strip to the door frame; wherein the assembling end is connected to a buffer, and the buffer is pushed against a door panel;

the insertion portion extends along a transverse direction, one end of the body along the transverse direction is an assembling end, the accommodation groove and the insertion groove are connected along the transverse direction, and the accommodation groove is in an elongated strip shape;

two expansion members, each of which includes a main body and a supporting section, a tenon section, and a side section that extend from the main body, the supporting section extends in the same direction as the tenon section, the side section extends in a opposite direction to the supporting section and the tenon section, the supporting section is abutted against an outer lateral surface of the door frame, one end of the tenon section that is not connected to the main body includes a tenon convex portion, the door frame includes two elongated grooves which are respectively at two opposite ends thereof along the transverse direction, two opposite outer surfaces of the elongated groove each include an engaging groove, the tenon section extends into the elongated groove, the tenon convex portion is engaged in the engaging groove, so that the expansion members are locked to the two opposite sides of the door frame, an expansion groove is formed between the side section and the main body, and the first section of the fasteners is inserted in the expansion groove;

wherein the door frame includes two fasteners, the fasteners each include a first section, a second section, and a hook which are connected in sequence, extension directions of the first section and the second section are different, the first section is inserted into the elongated groove to fix the two fasteners on the two opposite sides of the door frame, and the hook is hooked to the wall.

* * * * *